SAURABH RAWAT

Computer Science & Engineering Student • Aspiring Software Development Engineer (SDE)

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EDUCATION

Undergraduate Record

Graphic Era Hill University — B.Tech in Computer Science & Engineering | **Academic Record: 9+ CGPA in 1st & 2nd Year** | `Currently Pursuing (Aug 2022 – Present)`

ISC Board (Class XII): Intermediate (89%)

ICSE Board (Class X): High School (94%)

TECHNICAL SKILLS

Languages:	Java, C++, C, Python, JavaScript, TypeScript, SQL
Core CS:	Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks
Web & Backend:	HTML5, CSS3, React, Next.js, Node.js, TypeScript, REST APIs
Databases:	PostgreSQL, SQLite, Prisma, Supabase/Neon
Tools:	Git, GitHub, VS Code, Vercel
Additional:	Machine Learning, Computer Vision, YOLOv8, FastAPI, IoT

SOFTWARE ENGINEERING PROJECTS

Kaarobarie POS Billing Software — *Full-Stack Restaurant Management & POS SaaS Platform*

- **Architected** a multi-tenant restaurant management SaaS with multi-franchise tenancy, real-time table occupancy tracking, and kitchen order ticket (KOT) workflows using React, TypeScript, Node.js, Express, Prisma ORM, and PostgreSQL.
- **Implemented** 7-tier hierarchical Role-Based Access Control (RBAC), secure JWT authentication, and transactional order settlement logic, ensuring complete multi-tenant database isolation and robust API error handling.
- **Integrated** real-time bi-directional Socket.IO event synchronization for live kitchen displays and developed an ESC/POS thermal printing spooler (58mm/80mm), eliminating order coordination delays between dining terminals and kitchen stations.

Railway Gangmen Safety System — *Embedded Safety Architecture & Hazard Telemetry Pipeline*

- **Developed** a safety-focused embedded monitoring and alerting system to safeguard track maintenance crews from oncoming train collision hazards using C++, Arduino Uno, proximity/vibration sensors, and RF transceivers.
- **Implemented** a low-latency multi-node telemetry pipeline with threshold-based sensor calibration and fail-safe signal broadcasting via C++ and Python serial communication listeners, enabling reliable real-time approach warning.

AI-Powered Missing Children Finder — *Distributed Surveillance Video Processing & Trajectory Graph Grid*

- **Engineered** a distributed video-processing and surveillance search pipeline for identifying missing individuals across CCTV camera feeds using Python, FastAPI, Ultralytics YOLOv8, InsightFace embeddings, and DeepSORT tracking.
- **Implemented** automated vehicle license-plate recognition using PaddleOCR alongside a graph traversal algorithm across camera-network junctions, forecasting probable movement trajectories and expediting search operations.

IoT-Based Landslide Detection System — *Geological Instability Monitoring & Early Warning Network*

- **Developed** an IoT geological monitoring system to detect slope displacement and imminent landslides using Embedded C++, MPU-6050 accelerometer/gyroscope sensors, soil moisture probes, and continuous anomaly threshold filtering.
- **Implemented** edge data transmission and automated multi-level alert dispatch (Advisory, Warning, Critical) via C++, Python, and REST endpoints to notify downstream monitoring interfaces upon threshold breaches.

AI-Based Pothole Detection System — *Edge Computer Vision & Geo-Spatial Road Defect Platform*

- **Developed** a road surface defect and pothole detection solution utilizing client-side edge computer vision with TensorFlow.js, a lightweight CNN model, HTML5 Canvas, and Web APIs, classifying surface anomalies directly on device.
- **Integrated** automated GPS geolocation stamping and real-time cloud dispatch into Firebase Realtime Database with transactional deduplication, logging coordinate data to assist preventative infrastructure maintenance.

LEADERSHIP & ACHIEVEMENTS

IEEE Student Branch, GEHU — **Organising Committee Member**
TechnIEEEks '25, SAARTHI '25 (24-Hr Hackathon), TechnIEEEks '26
Coordinated technical operations, workshop logistics, and cross-functional committee execution across multi-day sessions.

Technology Business Incubator, GEU — **Organising Committee Member**
Graphethon 2.0 (48-Hr), Graphethon 3.0 (72-Hr)
Managed participant intake operations and on-ground logistics for 500+ attendees during flagship national hackathons.

- **SELECTED** **National-Level Hackathons** — Selected in competitive development sprints, incl. **MINDCRAFT 2K26 (Top-40 Selection)**.
- **SEMIFINALIST** **Economic Times X GenAI Hackathon** — Semifinalist in nationwide generative AI engineering competition.
- **FINALIST** **PRAXIS 2.0 (Google Developer Groups)** — Finalist in national technical hackathon building scalable software solutions.
- **FINALIST** **Neural Nexus AI/ML Challenge** — Finalist at Grafest 2026 machine learning & vision challenge.